L4a: Lattice Trading Rules and Terminal Probabilities

CHEME 5660 Cornell University

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we use net present value to evaluate a long stock position, then calculate the probability of exceeding a target at a scheduled sale time.

Today's concepts

- Evaluate a stock trade using NPV: derive the scaled NPV and explain the effects of sale price, holding period, and benchmark.
- Compute terminal target probabilities: find the minimum up-move count needed to exceed a target and add the binomial probabilities.
- Extend the lattice to m possible outcomes per step: use branch counts to calculate node prices, probabilities, and state counts.

Lectures and Examples

Lecture: Lattice Trading Rules and Terminal Probabilities

Example: Binomial Lattice Terminal-Target Probabilities
What is the probability of exceeding a target scaled NPV? We estimate the lattice parameters from historical data and compare probabilities across targets.

Example: N-Ary Recombining Lattice Models
What changes when we allow more price movements per step? We estimate branch factors and probabilities, build the lattice, and examine its terminal outcomes.

Concept Review: What Trades, Where, and How

1. Security

A stock is an ownership claim. An ETF is an exchange-traded basket of assets.

Mutual funds are priced periodically, usually once each trading day.

2. Venue

An exchange matches buy and sell orders under a common set of market rules.

A market maker helps trading by posting a bid and an ask.

3. Instruction

A market order prioritizes execution, but not a specific price.

A limit order controls the acceptable price, but may not execute.

Execution matters: the order type and current quotes determine the cash flows in a complete trading rule.

The Exchange Matches Buyers and Sellers

BUYERS
submit buy orders

EXCHANGE
ranks and matches orders

SELLERS
submit sell orders

TOP OF BOOK

BEST BID
$b_t = \$100.95$
highest current buy price

spread
$0.10

BEST ASK
$a_t = \$101.05$
lowest current sell price

Market sell executes at the bid.

Market buy executes at the ask.

Assume enough size is available at each displayed quote; fees and slippage are omitted.

Market Makers Supply Executable Prices

Citadel Securities is a market maker: it quotes prices at which it is willing to buy and sell, giving incoming orders an available counterparty.

One-share example using the preceding quotes

Market buy

$$\text{cash flow} = -a_t = -\$101.05$$

The trader receives one share and pays the ask.

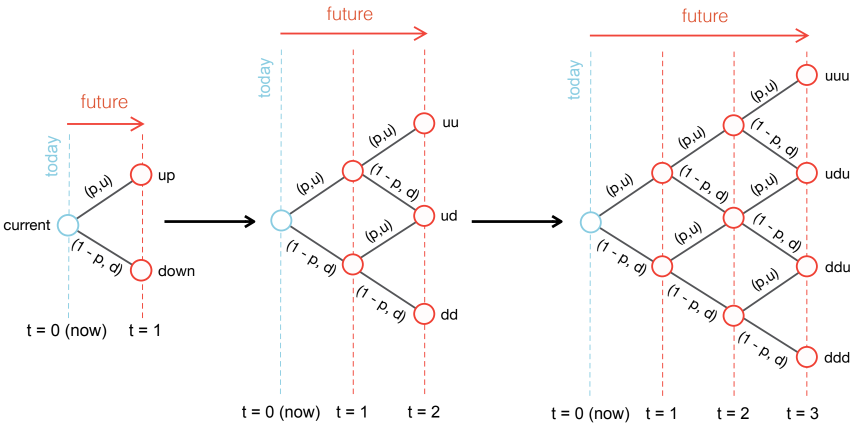
Market sell

$$\text{cash flow} = +b_t = +\$100.95$$

The trader delivers one share and receives the bid.

Modeling consequence: a complete trading rule uses executable entry and exit prices, plus any fees and slippage.

A Recombining Price Lattice



1-step ahead prediction

2-step ahead prediction

3-step ahead prediction

The Binomial Lattice Model

Let $S_0 > 0$ be today's share price. At each step, we flip a coin: the price changes by a factor u with probability p, or d with probability $1 - p$, where $u > d > 0$ and $0 < p < 1$.

We assume independent flips and hold (u, d, p) fixed. If K_t counts the up moves after t steps, then $K_t \sim \text{Binomial}(t, p)$:

$$\underbrace{S_{t,k} = S_0 u^k d^{t-k}}_{\text{node price}}, \qquad \underbrace{\mathbb{P}(K_t = k) = \binom{t}{k} p^k (1-p)^{t-k}}_{\text{node probability}}.$$

Each level contains $t + 1$ price states. Independent moves with fixed parameters do not capture volatility clustering.

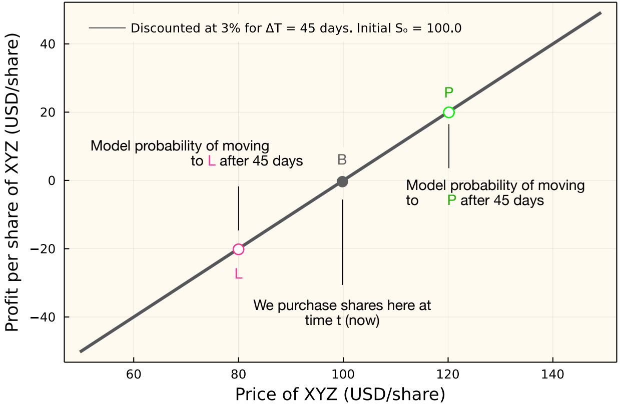
The Trade We Will Evaluate

Suppose we buy shares now and sell them after a chosen holding period. Our lattice supplies the possible sale prices and their probabilities.

- Buy $n_0 > 0$ shares at price S_0 at time zero.
- Hold for N steps of duration Δt years; sell at $T = N\Delta t$.
- Discount the sale proceeds using the annual risk-free benchmark g_y.
- Compare the discounted return with a chosen target $\rho_\star$.

We assume no dividends, transaction fees, or bid–ask spread. The optional execution-aware example adds trading costs.

Sale Price and Discounted NPV



Net Present Value of a Long Position

Buying n_0 shares costs $n_0 S_0$ today. Selling at time T returns $n_0 S_T$. Following L1b, g_y is the constant, continuously compounded annual risk-free growth rate associated with benchmark yield y.

$$\underbrace{\mathrm{NPV}(g_y, T)}_{\text{time-0 dollars}} = \underbrace{-\, n_0 S_0}_{\text{purchase today}} + \underbrace{n_0 S_T e^{-g_y T}}_{\text{present value of sale proceeds}}.$$

Divide by the initial investment to obtain the dimensionless scaled NPV:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\mathrm{NPV}(g_y, T)}{n_0 S_0} = e^{-g_y T} \frac{S_T}{S_0} - 1.$$

If $\rho_T > 0$, the discounted sale proceeds exceed the purchase cost. Scaling removes the dependence on the number of shares.

Short Holding Period

When $|g_y|T$ is small, $e^{-g_y T} \approx 1$. The scaled NPV is then approximately the fractional change in share price:

$$\underbrace{\rho_T}_{\text{scaled NPV}} \approx \underbrace{\frac{S_T}{S_0} - 1}_{\text{price return}} .$$

Under the binomial lattice, $K_N \sim \mathrm{Binomial}(N, p)$ and:

$$\frac{S_T}{S_0} = u^{K_N} d^{N-K_N}.$$

Therefore the $N+1$ terminal nodes give the complete discrete distribution of the approximate return.

Longer Holding Period

Retain the discount factor whenever $|g_y|T$ is not small. Which sale price makes the scaled NPV zero?

$$e^{-g_y T}\frac{S_T}{S_0} - 1 = 0 \quad \Longrightarrow \quad \underbrace{S_T = S_0 e^{g_y T}}_{\text{break-even sale price}} .$$

For $g_y > 0$, the price must grow beyond the benchmark to give a positive NPV. At a terminal node with k up moves, the exact scaled NPV is:

$$\rho(k) = u^k d^{N-k} e^{-g_y N\Delta t} - 1, \qquad k = 0, \dots, N.$$

We use this exact expression for all holding periods. The benchmark changes the return attached to a node, but not its real-world probability.

Cumulative Probability of a Terminal Target

Let $\rho_\star > -1$ be our target scaled NPV. At the scheduled sale time, we ask for:

$$\mathbb{P}(\rho_T > \rho_\star), \qquad \rho_T = \rho(K_N).$$

For example, $\rho_\star = 0.05$ requires the discounted sale proceeds to exceed the initial investment by more than 5%. Equality does not count.

Increasing the up-move count replaces one factor d with u. The price ratio increases by $u/d > 1$, so the scaled NPV increases as well.

Once one count exceeds the target, every larger count does too. We therefore solve for a threshold $\tau(\rho_\star)$ such that $K_N > \tau(\rho_\star)$.

Solve the Return Inequality for the Up Count

Begin with the strict target at a node containing k up moves:

$$e^{-g_y N\Delta t} u^k d^{N-k} - 1 > \rho_\star.$$

Move positive terms, take logarithms, and use $\log(u/d) > 0$:

$$k \log(u/d) > \log(1+\rho_\star) + g_y N\Delta t - N \log d.$$

Thus the real-valued threshold is:

$$\underbrace{\tau(\rho_\star)}_{\text{count threshold}} = \frac{\log(1+\rho_\star) + g_y N\Delta t - N \log d}{\log(u/d)}.$$

Strict Targets Need the Next Integer

The event is $K_N > \tau(\rho_\star)$. Since the up-move count is an integer, the smallest integer strictly above the threshold is:

$$\kappa = \lfloor \tau(\rho_\star) \rfloor + 1.$$

If $\tau = 2.4$, we need at least three up moves. If $\tau = 2$ exactly, we still need three: two would only reach the target.

There are only $N + 1$ feasible counts, from zero through N. We store:

$$k_{\min} = \min\{\max\{\kappa, 0\}, N + 1\}.$$

The values 0 and $N + 1$ indicate that all or none of the terminal nodes succeed.

Add the Successful Node Probabilities

The stored cutoff gives the terminal target probability:

$$\mathbb{P}(\rho_T > \rho_\star) = \begin{cases} 1, & k_{\min} = 0, \\ \displaystyle\sum_{k=k_{\min}}^{N} \binom{N}{k} p^k (1-p)^{N-k}, & 1 \le k_{\min} \le N, \\ 0, & k_{\min} = N+1. \end{cases}$$

For $\rho_\star \le -1$, the probability is one because every model price is positive. The logarithmic threshold is unnecessary.

In the code: we check terminal-node returns directly to identify $k_{\min}$, avoiding floating-point rounding of an equality boundary. For an interior cutoff, evaluate the complementary binomial CDF at $k_{\min} - 1$.

A Three-Step Target Calculation

Suppose $N = 3$, $p = 0.6$, and the chosen target requires at least two up moves. The successful outcomes are $K_3 = 2$ and $K_3 = 3$.

$$\mathbb{P}(K_3 \geq 2) = \binom{3}{2}(0.6)^2(0.4) + \binom{3}{3}(0.6)^3$$

$$= 0.432 + 0.216 = 0.648.$$

The model assigns a 64.8% probability to exceeding the target.

The return inequality determines which counts succeed. The up-move probability p determines how much probability those counts carry.

What About Falling Short of the Target?

At the scheduled sale time, the scaled NPV either exceeds the target or finishes at or below it. These events are complementary:

$$\mathbb{P}(\rho_T \leq \rho_\star) = 1 - \mathbb{P}(\rho_T > \rho_\star).$$

In our three-step example, the probability of finishing at or below the target is $1 - 0.648 = 0.352$.

Increasing g_y with the lattice and target fixed cannot increase the success probability. Changing N changes both the cutoff and the count distribution.

Scheduled sale: this calculation checks the final outcome. Selling when a boundary is first reached requires the price path; see the optional first-passage example.

A Three-Outcome Price Model

Suppose $S_0 = 100$ USD/share. At each step, the price can increase by 10%, remain unchanged, or decrease by 10%.

Movement	Factor f_j	Probability p_j	Price after one step
Up 10%	1.10	0.25	110
Unchanged	1.00	0.50	100
Down 10%	0.90	0.25	90

Up then down gives $100(1.10)(0.90) = 99$ USD/share. Reversing the order gives the same price. Both paths have counts $(1, 0, 1)$, so they share one node.

Nine Paths Recombine into Six Count States

After two steps, there are $3^2 = 9$ ordered paths. We only need six count states:

Counts (up, unchanged, down)	Paths	Price (USD/share)	Probability
$(2, 0, 0)$	1	121	0.0625
$(1, 1, 0)$	2	110	0.2500
$(1, 0, 1)$	2	99	0.1250
$(0, 2, 0)$	1	100	0.2500
$(0, 1, 1)$	2	90	0.2500
$(0, 0, 2)$	1	81	0.0625

For $(1, 0, 1)$, add the up–down and down–up probabilities:
$0.25(0.25) + 0.25(0.25) = 0.125$. All six node probabilities sum to one.

N-Ary Lattice Models

An N-ary lattice allows $m \geq 2$ possible price movements per step. Branch j has a positive factor f_j and probability $p_j > 0$, with $\sum_{j=1}^{m} p_j = 1$. We keep N for the holding-period step count.

After t steps, the branch-count vector is:

$$\mathbf{x} = (x_1, \dots, x_m), \qquad x_j \in \mathbb{Z}_{\geq 0}, \qquad \sum_{j=1}^{m} x_j = t.$$

Each occurrence of branch j multiplies the price by f_j. Fixed factors make the final price depend on the counts, while the intermediate prices can depend on the order.

N-Ary Node Values

For count vector $\mathbf{x}$ at level t, the price is:

$$\underbrace{S_t(\mathbf{x})}_{\text{node price}} = S_0 \prod_{j=1}^{m} f_j^{x_j}.$$

Assume independent branch choices with fixed probabilities. If $\mathbf{X}_t$ is the random count vector after t steps, then:

$$\mathbb{P}(\mathbf{X}_t = \mathbf{x}) = \underbrace{\frac{t!}{x_1! \cdots x_m!}}_{\text{number of orderings}} \underbrace{\prod_{j=1}^{m} p_j^{x_j}}_{\text{probability of one ordering}}.$$

For $m = 2$, this reduces to the binomial probability $\binom{t}{k} p^k (1-p)^{t-k}$.

How Many States Do We Need?

A count state satisfies $x_1 + \cdots + x_m = t$. Represent the t moves by stars and separate the m branch groups using $m - 1$ bars:

$$\star \mid \mid \star \quad \longleftrightarrow \quad (1, 0, 1).$$

Choose the bar positions among $t + m - 1$ available positions:

$$L_t = \binom{t + m - 1}{m - 1}, \qquad L_2 = \binom{4}{2} = 6 \quad \text{when } m = 3.$$

This counts branch-count states, not necessarily distinct prices. If $f_1 f_3 = f_2^2$, for example, $(1, 0, 1)$ and $(0, 2, 0)$ share a price. We add their probabilities when computing the probability of that price.

N-Ary Example: Historical Branch Estimates

How do we construct a lattice from historical observations? In the companion example, we work through the calculation for a selected firm.

1. Group historical growth observations into bins.
2. Average the price-change factors within each bin and use observation frequencies to estimate branch probabilities.
3. Construct node prices from the branch counts.
4. Calculate multinomial probabilities and check that they sum to one.
5. Examine the terminal share-price distribution and its probabilities.

Calibration matters: averaging within bins removes within-bin variability. A valid probability sum does not show that the branches preserve the historical distribution.

Optional Advanced Material

Advanced: First-Passage Exit Rules
What changes if we check the price after every step? We track positions until a take-profit or stop-loss boundary is first reached, then compare with checking only the final price.

Advanced: Execution-Aware Probability of Profit
How do trading costs change the target probability? We include the bid–ask spread, fees, and unfavorable execution prices (slippage), then compare the result with the frictionless calculation.

Optional; neither notebook is a prerequisite for L4b.

Optional: Terminal and First-Passage Rules

A terminal rule checks the trade at the scheduled sale date T. A first-passage rule sells when a take-profit or stop-loss boundary is first reached.

- Paths can end at the same price but cross a boundary at different earlier times. Their trading outcomes can therefore differ.
- At each step, track positions that remain open and record the probability of reaching either boundary first.
- Compare those results with the probability of crossing neither boundary before the holding period ends.

Terminal node probabilities alone cannot determine when a boundary was reached.

Summary

We used NPV to evaluate a long stock position, computed terminal target probabilities, and extended the binomial model to several branches.

Key takeaways

- NPV-based trading framework: We scaled the NPV by the initial investment to compare discounted sale proceeds with the purchase cost.
- Terminal target probabilities: We found the minimum up-move count needed to exceed a target and added the probabilities of the successful counts. The complement gives the probability of finishing at or below it.
- N-ary lattice models: We used branch counts to calculate prices, multinomial probabilities, and state counts when each step allows m outcomes. More branches also increase the computational cost.

Next time: continuous stochastic models of price dynamics.

Next time: continuous stochastic models of price dynamics.